

Surface potential decay of functional dielectrics after polarization

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High voltage polarization domains in functional dielectrics and the ability to orient them with an external applied electric field lead to the development of a variety of applications of energy conversion from energy harvesting to information storage and reading. Voids are often generated at the bond interface during the production process of polymer films, which can break down easily in the process of polarization with high voltage, but the voids inside dielectric materials play an important role in charge storage and maintenance if the materials are oriented to the application of electrostatic generator, sensor, actuator and transducer etc. Meanwhile, a polymer matrix doped with a small amount of nanoparticles can suppress charge formation and increase surface charge decay rate. Results obtained in this paper show that polymer materials with microvoids have better maintenance properties of space charge compared with nanocomposite polymer materials. The research result is of important reference value and practical significance in the field of insulation and applications of functional dielectric materials.

Statement of Novelty

The originality of this paper lies in the comparison of the surface potential decay characteristics of cellular polypropylene films and silicon nanofilms; measurement of their surface potentials after corona poling; and discovery of the fact that enhanced charge storage performance can also be achieved by using an effective corona poling method on cellular polymers and not only by adding micro- or nanoadditives. Last but not least, the authors found from the results that the charge storage stability of the voids inside polymer films is better than those in silicon films doped with nanoparticles, and they analyzed the cause of this phenomenon. The major contribution of this paper lies in designing and developing work to meet a variety of application requirements for electrets.

1. Introduction

The dynamic surface potential characteristic of functional dielectric materials, which is a significant sign of charge storage stability, is the most important property of electrets.^{1,2} While the phenomenon that the charge storage ability of cellular polypropylene (PP) is better than that of poly(vinyl chloride) (PVC) has been studied,³ further study of the migration of material charge and the suppression of internal charge is necessary.⁴⁻⁷ Moreover, the special structures of functional dielectrics, such as voids and molecules, play an important role in the surface charge decay properties of a material. The effects of the surface roughness of a polymer film on the surface

charge decay property has been investigated.⁸ The direct impacts of the thickness of film on charge density, internal electric fields, bulk conductivity and migration rate of detrapped charge stored in electret films have been discussed,^{9,10} and the influences of thickness on the charge storage stability of material and charge retain capability are also analyzed. Kojima *et al.*¹¹ analyzed the distribution of polymer surface charge under vacuum and atmospheric conditions. But the comparison and the analysis of surface charge properties of functional dielectric materials with microvoid structure and nanoparticles have not been reported in literature, even though these properties are essential to materials in the application of insulation materials and functional dielectrics.¹²⁻¹⁴ Xu *et al.*¹⁵ tested the surface charge of corona-charged polyethylene and found that the charge injected in a deep trap leads to slow decay of surface potential. The present paper systematically discusses the influence of the surface charge decay properties of the cellular PP electret films and the silicon polymer films embedded with nanoparticle. Also, theoretical and technical support for the research and development of new electret materials for sensors and actuators and the study of the internal charge accumulation effect are discussed in this paper.

2. Experimental

2.1 Samples

PP and silicone rubber in practical engineering are widely used in insulation parts of high-voltage electrical equipments. Voids will

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appear in the process of PP production. However, the surface charge will accumulate in the void. This feature is unfavorable for the insulation of the material, but it is advantageous to functional dielectric materials.¹⁶ In the same way, because surface charge also accumulates on the surface of the added nanoparticles, magnetic iron (II,III) oxide (Fe_3O_4) nanoparticles can enhance material storage capacity.¹⁷ To study the material charge storage ability, samples of silicon films are doped with iron (II,III) oxide nanoparticles.

The specimens are cellular PP with a thickness of 50 μm and an area of 40 mm $\times$ 40 mm. The production process of cellular electrets is same as that for the sample made by Tang *et al.*¹⁸ And silicon films doped with 2% weight of iron (II,III) oxide nanoparticles in the production process (silicone rubber +2% iron (II,III) oxide) is used in the experiment. Three samples are prepared and tested in the experiments. All the samples are vacuum evaporated with a 50-nm-thick gold (Au) electrode on one side. This makes the specimens a type of a film with a grounded electrode, which makes testing and analysis after corona discharge easy.

2.2 Direct current negative corona discharge system

The grid-control negative corona discharge system in this experiment consists of a high-voltage generator with a permanent direct current source, a needle electrode, a metal grid, and samples. The time for corona discharge is 10 min, and the discharge temperature is room temperature. The voltage of needle electrode and the grid are -30 and -4 kV respectively. The distance between the needle electrode and the specimen is 4 cm. A metal grid is placed between the needle electrode and the specimen and kept at a safe distance of 2 cm from the specimen. These requirements are used to ensure the even injection of electrons into the specimens and the uniform distribution of sample surface charge.¹⁹

2.3 Measurement and analysis system of dynamic surface potential

Surface charge measurement system is shown in Figure 1. This system consists of a digital oscilloscope, a non-contact

electrostatic voltmeter, electrostatic probes and samples. The oscilloscope simultaneously records the surface charge decay of the tested samples, which is converted into standard surface charge decay rates. Measuring and comparing the surface decay rates of the different samples can reflect the decay property of the surface charge of each sample. Meanwhile, in order to compare the surface charge decay rates of specimen objectively, accurately and systematically, two measuring time spans are applied during the experiment, which are continuous measurement after discharge and within 24 h.

3. Results and discussion

The decay characteristic of surface charge is an important quality index of electret charge storage stability. In Figure 2, the two curves show the surface charge decay characteristics of the cellular PP film and the silicon film doped with 2% iron (II,III) oxide nanoparticle respectively. The measuring time starts after discharge and ends within 24 h.

For the same measuring time, by comparing the normalized surface potential of the cellular PP film and the silicon film doped with 2% iron (II,III) oxide nanoparticle, it is found that charge decay rate of the electret film with voids is significantly higher than that of the insulation silicon dielectric film doped with nanoparticles. Four hours after grid-control negative discharge, the equivalent surface potentials of the silicon film doped with nanoparticles and the cellular PP film have both shown low values, while the surface charge of the cellular PP dielectric materials remains at a relatively high potential level of the absolute terms for more than 3–2 kV, which is equal to 80% of the initial value. In addition, for the silicon film doped with 2% nanoparticles, surface potential is relatively stable from 1 to 2 h, while the stable period of cellular PP film surface potential relatively lags behind. And the surface potential is stable in two separate periods, which are from 5 to 8 h and after 22 h.

Inorganic nanoparticles and polymer films have different chemical and dielectric properties. Charge transport will occur between them when the nanoparticles are mixed in the polymer film, and

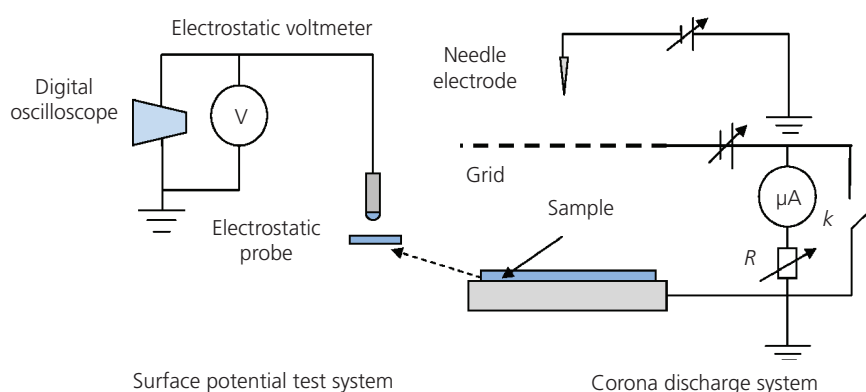


Figure 1. Surface charge measurement system

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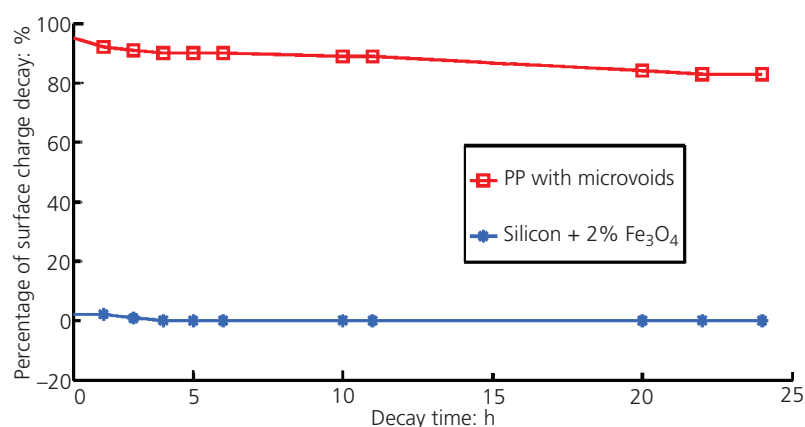


Figure 2. The potential decay rate after polarization (0–24 h)

nanoparticles take a positive electric charge, while the polymer takes a negative electric charge. Therefore, the added nanoparticles will weaken the electric field intensity of the electrode and then will suppress the charge injection from the electrodes to the polymer film. Meanwhile, they can also suppress the generation and development of electricity branches. In consequence, added nanoparticles in the polymer can improve the electric strength and insulation performance of the dielectric materials.²⁰

During charging, space charge plays an important role in polymer degradation. Due to the high-energy field, an electron achieves high kinetic energy and is able to break polymer bonds. Thus, the space charge is injected into the polymer bulk. Then C–H bonds of cellular PP are broken and continually combine with OH and OOH groups as shown in Figure 3. Such highly energetic ion avalanches generated by discharge in a gas filled with microvoids degrade the insulation and break the chemical bonds which constitute the polymer chains.^{21,22}

Accordingly, more charges are accelerated by the ion avalanches and injected into the polymer matrix. Thus, the charge storage stability of the cellular PP is better than that of the polymer film doped with nanoparticles.

In addition, the test results show that voids inside the functional dielectric film could be effective in charge storage. During the corona polarization process, Paschen breakdown occurs in a single void, so the internal upper and lower walls of the void constitute the tiny unit of charge storage; the number of charge traps and charge detrapping energy increase. Therefore, this leads to a rise in charge potential not

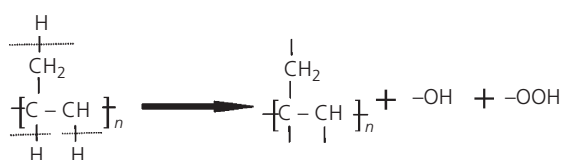


Figure 3. Change in chemical bonds of cellular PP

only inside the dielectrics but also on the surface of the dielectrics, which keeps a high potential level in a long period of time.²³ The nanoparticle structure in the polymer film is shown in Figure 4. Meanwhile, the nanoparticles in polymer matrix not only make some contribution to the increase in charge density but also cause the rise in the conductivity of the functional dielectric materials, which leads to a significant decline in internal storage charge density.²⁴ Kumar *et al.*²⁵ worked on pure and carbon-coated lithium manganese phosphate (LiMnPO₄) nanorods by impedance measurements, and the conductivity values evaluated from impedance results confirmed that more than two orders of magnitude of conductivity enhancement was observed in the carbon-coated lithium manganese phosphate nanorods compared with the pure ones. This is the reason why the surface potential of the polymer film with nanoparticles shows a faster decay, and it also explains why the surface potential stable period is prior to that of the cellular PP electret film.

Recently, some scholars have proved that the charge trap memory of a multilayer film is a kind of trap charge storage structure which has extreme important application prospects. For instance, Tang *et al.*²⁶ used a multilayer hafnium (IV) oxide (HfO₂)/aluminum oxide (Al₂O₃) film as the storage layer and made a systematic analysis of trap charge storage devices. The results show that the multilayer storage structure allows more charge storage in the interface traps between hafnium (IV) oxide and aluminum oxide. Bhuyian *et al.*²⁷ found a fact that the increased coupling between the electrons trapped at adjacent oxygen vacancy defect could reduce the defect activation energy. Thus, the charge storage performance is different between cellular dielectrics and functional materials with nanoparticle structure. The purposes of these studies are to investigate the charge storage ability, improve the energy conversion properties of dielectric materials and to broaden the application field of dielectric materials.

4. Conclusions

This paper shows the effects of the void structure and the nanoparticle structure inside functional dielectric materials on the number of charge traps and the surface charge stability by

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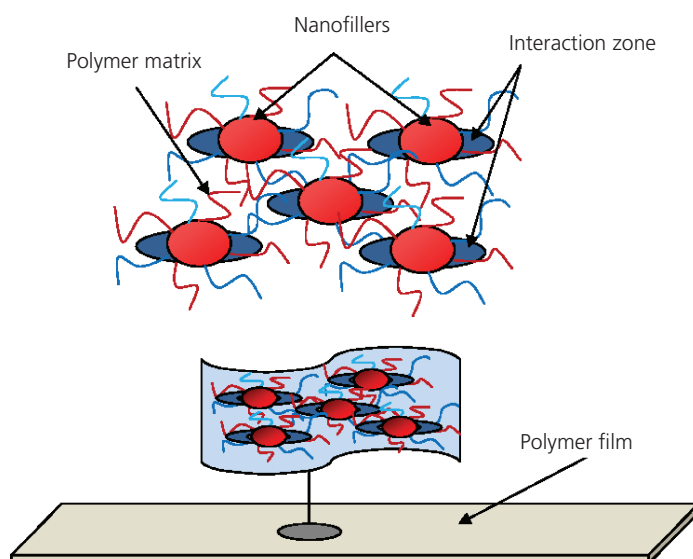


Figure 4. Schematic diagram of nanoparticle structure in polymer film

measuring the dynamic surface potential characteristics of materials with different microstructures. The results demonstrate that the voids inside dielectric materials play an important role in charge storage and maintenance. At the same time, a polymer matrix doped with a small amount of nanoparticles can suppress charge from forming and increase surface charge decay rate. Consequently, it can be found from the results that the charge storage stability of voids inside a polymer film is better than that of voids inside a silicon film doped with nanoparticles.

An important theoretical and experimental basis for designing and developing work to meet a variety of application requirements for space charge electrets is provided in this paper. These conclusions are essential from the aspects of applications of insulation and functional dielectric materials. Based on the result, it is established that enhanced charge storage performance can also be achieved by using an effective corona poling method on cellular polymer and not only by adding micro- or nanoadditives.

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